



Systems Engineering Maintenance and Reliability Theory

Prof. Dr. Kai Höfig



Reliability, Availability, Safety

- ◆ **Reliability** is a quality attribute of a system giving the **probability** for a certain point in time, that the system's functions are behaving as specified.
- ◆ **Availability** is a quality attribute of a system giving the **fraction** of the lifetime of a system where the primer system's functions can be used as specified.
- ◆ **Safety** is a quality attribute of a system. A system is safe, if no unacceptable risks can emerge from the system, what is **not quantified** in general.
- ◆ *Those quality attributes are at war: A safe stop of an escalator makes the system safer but also decreases the availability.*



Maintenance

- ♦ **Maintenance** is the **process** of executing **tasks** on a system in operation to maintain reliability, availability or safety, depending on what is important for its operation.
- ♦ **Preventive Maintenance** are all maintenance activities that aim at preventing failure.

Definition used here,
there are many
different out there!



The opposite of preventive maintenance is run-to-failure

- ♦ **Run-to-failure** means operating equipment until a failure occurs. **Repair** is then required (a corrective/reactive maintenance task).
- ♦ **Preventive maintenance (PM)** means acting before a failure occurs and replace **repair** with other maintenance activities. PM has two central goals:
 1. **Save costs** by replacing costs for repairs with costs for preventive maintenance.
 2. **Reduce risk** of death, injury or environmental damage to an acceptable level



A typical run-to-failure component



Maintenance activity to **reduce risk**. Replacing a tire is safer, but costs more than run-to-failure.



Maintenance activity to **reduce costs**. Replacing engine oil is cheaper than letting the engine fail.



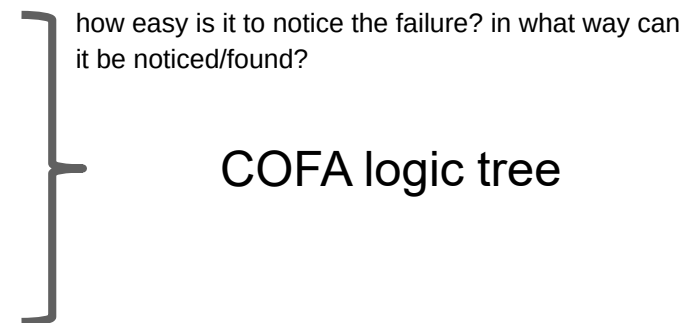
Maintenance strategies examples

- ◆ The selection of the maintenance strategy of a component depends on
 1. the risk on system level for a failure of that component
 2. and on the nature of how the failure occurs or is observed.
- ◆ Replace components before they reach their lifetime.
- ◆ Lube components frequently.
- ◆ Measure oil level and refill when low
- ◆ Measure thickness of a brake disc and replace when too low (condition directed PM task).
- ◆ Monitor vibration pattern that indicates wear of a bearing and predict residual lifetime using AI.
- ◆ Let a redundant generator run till failure and then replace or repair it. In this case, there is no functional failure on system level, but only on component level.



Component Classification

- ◆ A system (or so-called equipment under control) needs to be decomposed until a level of manageable parts is reached. The entity of this level of decomposition is here called an **item**.
- ◆ A COFA (Consequence of failure analysis) is performed on that level of decomposition to identify
 - the functions of an item what does it do
 - the functional failures of an item and the dominant failure modes in what ways can it fail?
 - the recognizability of the failure mode
 - effect on system level (hazard)
 - consequence on system level (harm) depending on the quality criteria
 - **component classification**





Maintenance task strategies (1)

- ◆ **Reactive maintenance activities**, performed after a failure has occurred.
 - low-cost components
 - low failure consequences
 - failures that are immediately evident

- ◆ **Preventive maintenance**, tasks performed after fixed time or usage intervals. The goal is to reduce failure probability.
 - wear-out components
 - legally required inspections
 - components with known lifetime behavior

- ◆ **Corrective maintenance**, performed after a fault, but before lost of function.
 - components with detectable faults
 - situations where immediate shutdown is not required
 - coordination with operational schedules



Maintenance task strategies (2)

- ◆ **Condition-based maintenance**, triggered by the observed condition of a component. Requires inspections, measurements, or monitoring.
 - components with progressive degradation
 - systems where early warning signs exist
 - avoidance of unnecessary preventive work

- ◆ **Predictive Maintenance**, advanced form of condition-based maintenance using data analysis, trends, and models to predict remaining useful life.
 - critical components
 - sensor-equipped systems
 - components with measurable degradation patterns



Run-to-failure components

- ◆ A **run-to-failure component** is not a safety-critical component or something that is critical for the availability of the system. Its failure is evident during normal operation and replacing a damaged run-to-failure component is neither too expensive nor does it result in long downtime.
- ◆ **Reactive maintenance activities**, performed after a failure has occurred.
- ◆ Classifying something as such a component requires knowledge about
 - Its failure impact on the system
 - Its failure costs
 - Its failure frequency and repair time to decide whether this is acceptable
- ◆ Examples: *normally on* indicator lights on a control panel, single plant lightings, toilets if redundant.



Economic components

- ◆ An **economic component** is not a safety-critical component or something that is critical for the availability of the system. Its failure is evident during normal operation and replacing a damaged economic component can be more costly than keeping it alive.
- ◆ **Preventive maintenance**, tasks performed after fixed time or usage intervals.
- ◆ **Corrective maintenance**, performed after a fault, but before lost of function.
- ◆ Classifying something as such a component requires knowledge about
 - Its failure impact on the system
 - Its failure costs
 - Its failure frequency and repair time to decide weather this is acceptable
- ◆ Examples: Conveyor belt drive motor in a warehouse, HVAC unit in a commercial building, Industrial air compressor, Pump in a non-critical fluid transport system



Commitment components

- ◆ A **commitment component** has failures that are not directly evident to the personnel and do not affect quality of service. These components usually fulfill external commitments and legal obligations.
- ◆ **Preventive maintenance**, tasks performed after fixed time or usage intervals.
- ◆ Classifying something as such a component requires knowledge about
 - Its failure impact on the system
- ◆ Examples: Fire detection and alarm system in a public building, elevator safety devices, emission monitoring sensors

failure of component is not expensive but due to external or legal obligations (commitment) the maintenance of the component must be upkeep



Potentially critical components

- ◆ A **potentially critical component** has failures that are directly evident to the personnel but do not *directly* affect safety or central functionality under normal conditions.
- ◆ **Preventive maintenance**, tasks performed after fixed time or usage intervals.
- ◆ **Condition-based maintenance**, triggered by the observed condition.
- ◆ Classifying something as such a component requires knowledge about
 - Its failure impact on the system under abnormal conditions
 - Its P/F interval potential failure to functional failure: how long until total failure occurs and the component does not work anymore
- ◆ Examples: Auxiliary hydraulic pump, cooling fan in an IT server or control cabinet, lubrication system filter

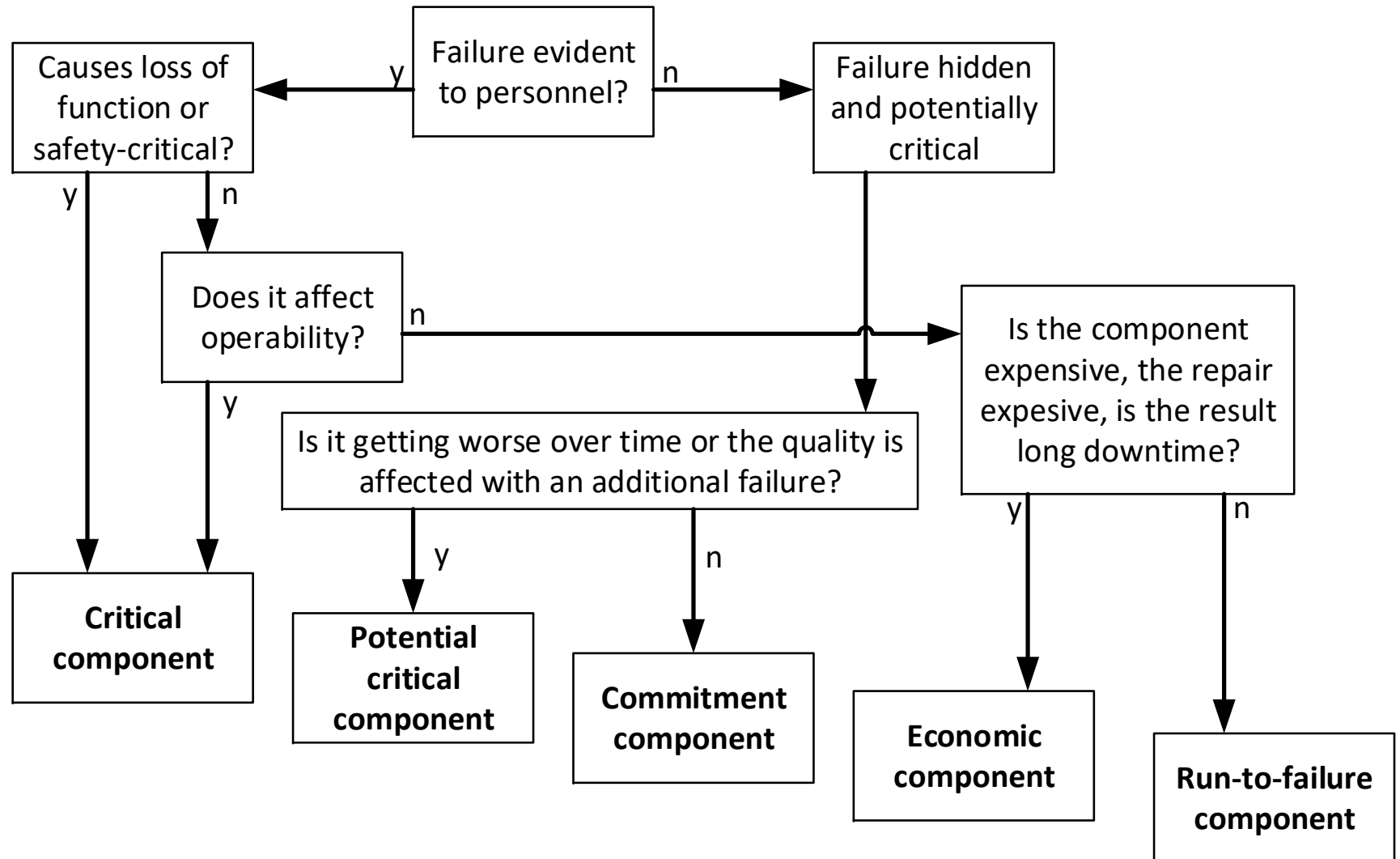


Critical components

- ◆ A **critical component** directly affects safety or central functions and needs to be proactively maintained.
- ◆ **Preventive maintenance**, tasks performed after fixed time or usage intervals.
- ◆ **Corrective maintenance**, performed after a fault, but before lost of function.
- ◆ **Condition-based maintenance**, triggered by the observed condition.
- ◆ **Predictive Maintenance**, advanced form of condition-based maintenance using data analysis, trends, and models to predict remaining useful life.
- ◆ Classifying something as such a component requires knowledge about
 - Its failure impact on the system
 - Its P/F interval
- ◆ Examples: Emergency shutdown valve in a chemical plant, aircraft flight control actuator, primary braking system in a vehicle or train, central PLC controlling a production line



COFA logic tree for component classification





COFA Worksheet example



Component	Functions	Failures	Dominant Failure Mode	Evident?	System Effect (Hazard)	Consequence on System Level (Harm)	Component Classification
2 way Isolation valve	Provide flow from pump to heat exchanger	fails to provide flow	stuck at closed	Yes	No cooling provided	Power reduction of 25% for 10h	critical
	Provide isolation when pump not working	fails to provide isolation	stuck at open	Yes	No system effect	No plant consequences	run-to-failure
	Provide emergency cooling to backup heat exchanger	fails to provide emergency cooling	fails to transfer to emergency position	Yes	No emergency cooling provided	Plant shutdown	critical

Classification?



Task planning

- ◆ **Reactive maintenance activities**, performed after a failure has occurred.
 - Always possible after a failure.
- ◆ **Preventive maintenance**, tasks performed after fixed time or usage intervals. The goal is to reduce failure probability.
 - What is the best time interval? → **Six Sigma**
- ◆ **Corrective maintenance**, performed after a fault, but before lost of function.
 - Fault detection possible? Which faults do not lead to failure?
- ◆ **Condition-based maintenance**, triggered by the observed condition of a component. Requires inspections, measurements, or monitoring.
 - Is the component degrading? Is the degrading long enough to take measures? → **P/F Interval**
- ◆ **Predictive Maintenance**, advanced form of condition-based maintenance using data analysis, trends, and models to predict remaining useful life.
 - Is the component degrading? Is the degrading long enough to take measures? → **P/F Interval**



Exercise – COFA Worksheet of an Automated Parking Garage

◆ System Description

The system under consideration is an **automated parking garage** belonging to a shopping center. The parking garage is open daily and is used by customers without on-site operating personnel. The system is monitored remotely.

The parking garage consists of:

- an entrance and exit with barrier systems,
- a payment and ticketing system,
- safety and monitoring equipment,
- technical infrastructure such as control systems and lighting.

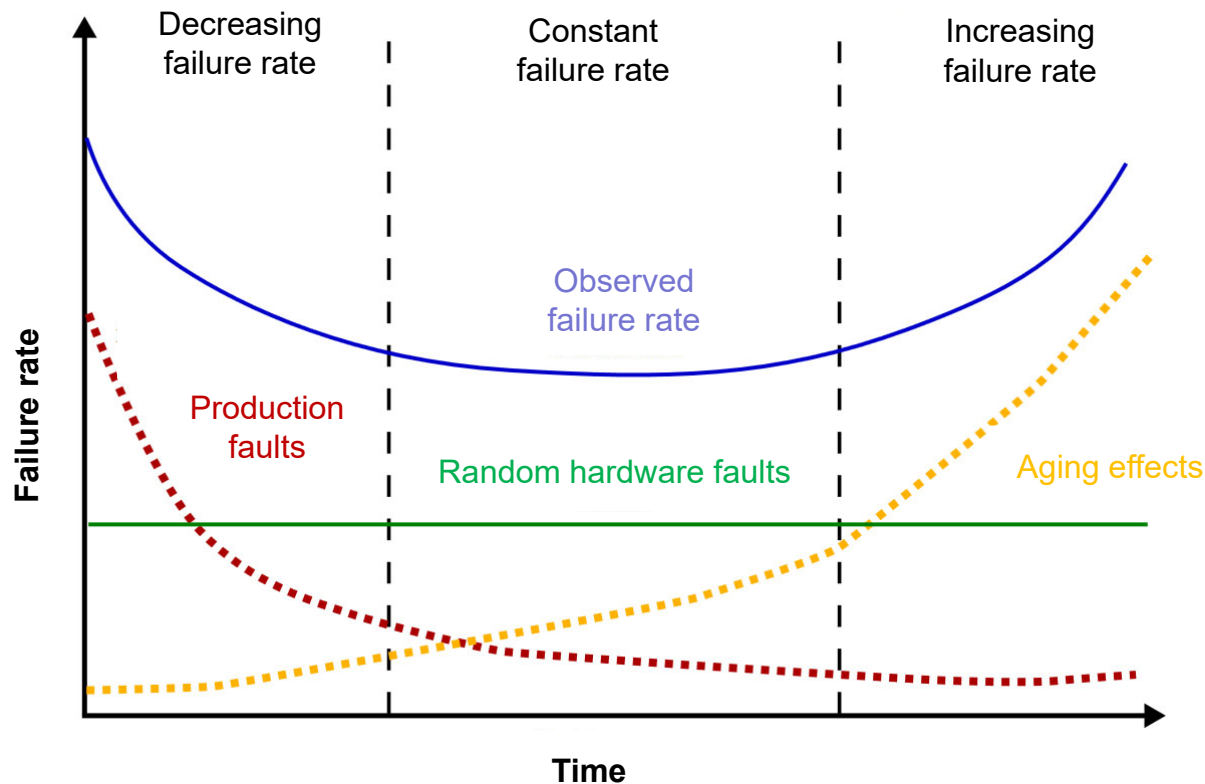
◆ Components to Be Analyzed

Analyze the following **eight components** of the system using the provided COFA Worksheet:

1. Barrier drive at the entrance
2. Central control unit / PLC of the parking garage
3. Ticket printer at the entrance
4. LED display "Parking available / full"
5. Fire detection system
6. Video surveillance camera
7. Lighting of individual parking levels
8. Ventilation system for exhaust gas removal



Failure Distribution



- ◆ The failure of hardware parts usually follows the bathtub-curve. At the beginning, faults in the production lead to failures during usage. After that follows a constant failure rate of random hardware faults and at the end of the lifetime, aging effects lead to more failures.



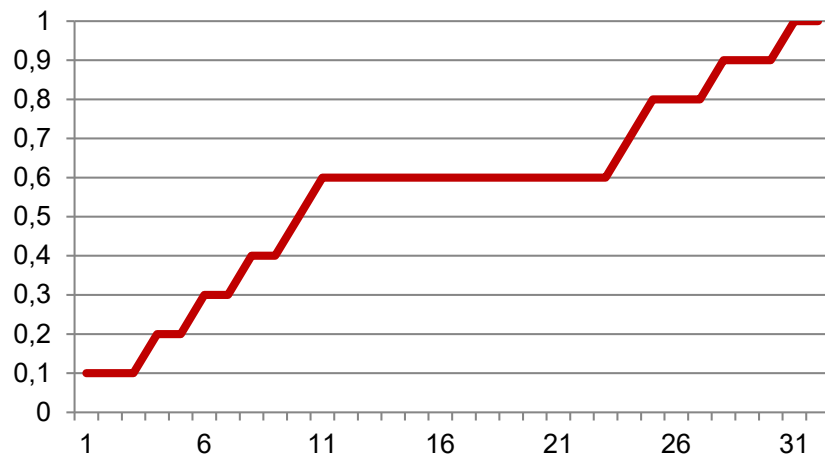
When the failure-rate is constant, the failure distribution is exponential

- ◆ Failures that are based on random faults (often electric), result in a exponential distribution of lifetime. Constant failure rates cannot be observed at the beginning or the end of a units lifetime.
- ◆ Example with a constant failure rate of 0,29
- ◆ In practical applications, this rate is either measured or estimated

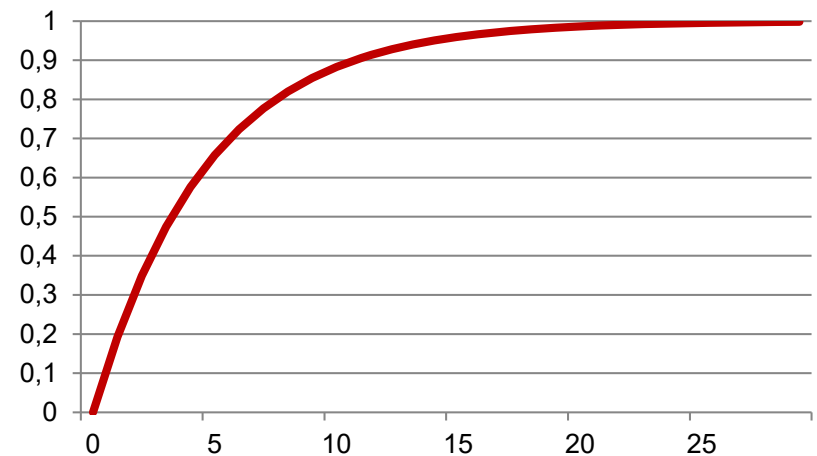
$$F(t) = \begin{cases} 1 - e^{-\lambda t} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

$$R(t) = 1 - F(t) = \begin{cases} e^{-\lambda t} & t \geq 0 \\ 1 & t < 0 \end{cases}$$

Empirical Distribution



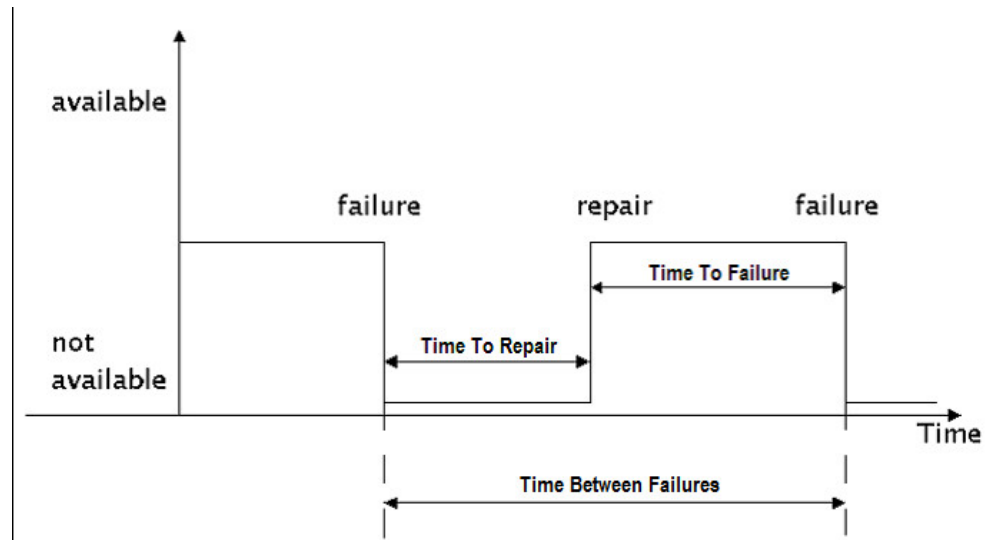
Exponential distributed lifetime with $\lambda=0,29$





Classic Reliability Figures – Does it look like that?

- ♦ MTTF – Mean Time To Failure
 - Mean time until a failure appears after the system has set into operation or after it has been restarted.
- ♦ MTTR – Mean Time To Repair
 - Mean time that it takes to repair a system
- ♦ MTBF – Mean Time Between Failure
 - Mean time between two failures, e.g. 1 billion hours. The inverse is the failure rate of a system, e.g. 1 failure per a billion hours



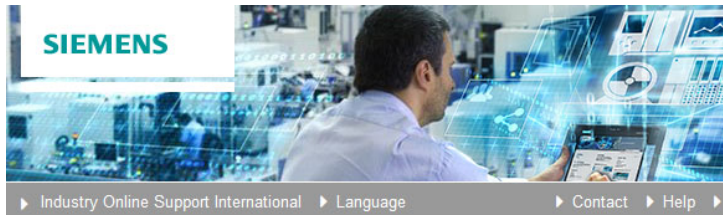
Risk Acceptance Criteria MEM

$$\begin{aligned} & 1 \cdot 10^{-5} \text{ per person and year} \\ &= \frac{1}{10^5} \cdot \frac{1}{8760h} \\ & MTBF = 10^5 \cdot 8760h = 876\,000\,000h \approx \frac{1}{1 \text{ FIT}} \end{aligned}$$

$$MTBF = MTTF + MTTR = \frac{1}{\lambda}$$



How do we get a failure-rate for our system?



Mean Time Between Failures (MTBF) - list for SIMATIC products

Entry Associated product(s)

MTBF = Mean Time Between Failures = time between two consecutive failures of a component.

This value which is given in years or hours (MTBF list contains values in years) is a pure statistical value. It is calculated for the modules of a component. The values given here are generally based on an ambient temperature of 40 degrees C.

- ◆ By **part** from standards (e.g. SN29500), aging tests, field data or estimates.

- ◆ By **product** often from the vendor

MTBF Calculator by ALD

Perform reliability prediction and MTBF/FR calculation for electronic and mechanical components in 5 simple steps:

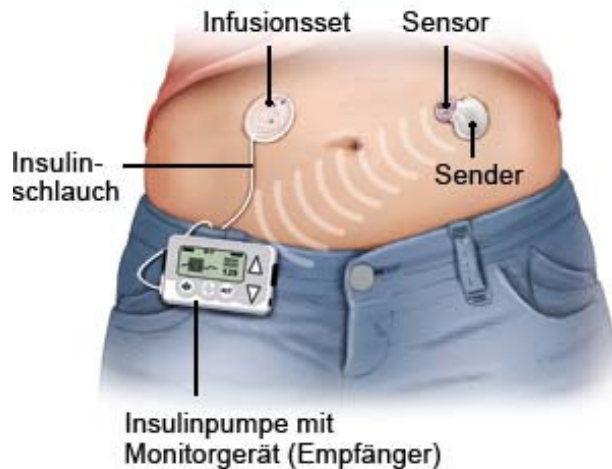
- 1. Select Component Family and Type**
Family: ELECTRONIC
Item Code: IC-Memory
- 2. Select Reliability Prediction Method**
MIL-217E-1 Part stress
MIL-217F-1 Part count
MIL-217F-1 Part stress
MIL-217F-2 Part count
MIL-217F-2 Part stress
ALCATEL
BELLCORE Issue 5
BELLCORE Issue 6
BRITISH TELECOM HRD4
BRITISH TELECOM HRD5
CNET RDF93 rev 02/95
- 3. Select Environment and Temperature**
Environment: GB Ground, Benign
Temperature: 25 degrees Centigrade
- 4. Enter Component Parameters**
Calculate
- 5. Get MTBF and FR**
MTBF: 12811394.41 hours
Failure Rate: 0.0781 failures per million hours
Failure Rate: 78.0555 FIT

ALD MTBF Calculator is a free tool suitable for simple reliability prediction of single components. If you need professional Reliability Tool for reliability engineering of complex systems, including product tree building, Reliability Block Diagrams, Reports, Report Generator, Pareto Analysis, Temperature Curve, Fault Tree Analysis, FMEA/FMECA, Safety Module, Derating Module and much more - please check our RAM Commander Software. You may download its evaluation version for free from our website. Copyright ALD Ltd. 2009 support@ald.co.uk www.aldservice.com

- ◆ For the **system** by adding up the failure rates (careful: they have to have the same base, e.g. *per year*). This approach is called *parts count*. This is a coarse pessimistic overestimation, since not every fault leads to a system failure or to a dangerous failure.



Example: Insulin Pump



- ◆ We are assuming, that an insulin pump can have a self-check that shuts down the device in case of a failure. (no safety impact)
- ◆ If in one year 5‰ of these devices have a failure (independently from aging effects or number of devices in use)

$$\lambda = 0.005 \text{ per year}$$

- ◆ The mean time till failure is

$$\text{MTTF} = 1/\lambda = 200 \text{ years}$$

- ◆ After 20 years approximately 10% of the devices are down:

$$F(20) = 1 - e^{-0.005 \cdot 20} \approx 0.0952$$



Preventive Maintenance Task Planning

- ◆ If we know, that a component fails exactly after 200 hours of operation, we will replace it after 199 hours and 59 minutes.
- ◆ That is not the case. Failures have a deviation. MTTF is just a mean.
- ◆ So statistically, the higher the deviation, the sooner we need to do replacement.

- ◆ Component A fails at 1, 10, 250 hours. MTTF is 87 hours

$$\sqrt{\frac{\sum (X - \bar{X})^2}{n}} = \sqrt{\frac{(1 - 87)^2 + (10 - 87)^2 + (250 - 87)^2}{3}} = 115$$

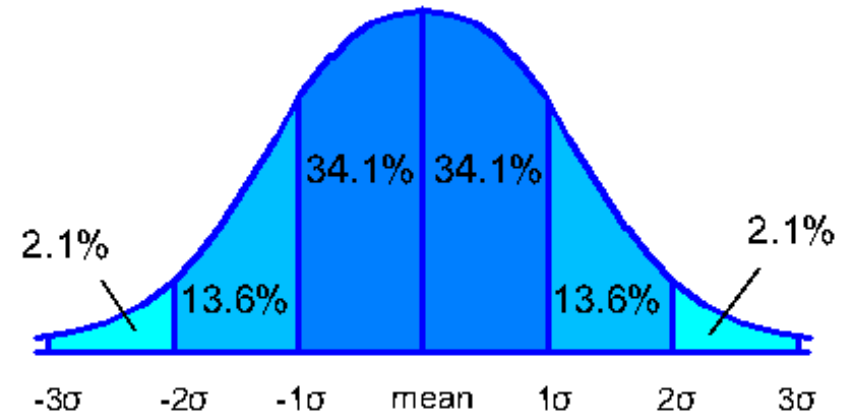
- ◆ Component B fails at 79, 89, 93 hours. MTTF is 87 hours

$$\sqrt{\frac{\sum (X - \bar{X})^2}{n}} = \sqrt{\frac{(79 - 87)^2 + (89 - 87)^2 + (93 - 87)^2}{3}} = 5.9$$



Six sigma

- ◆ If we plan to exchange the component at MTTF, our chances are 50/50 that we change before failure in standard normal distribution.
- ◆ If we exchange 1 standard deviation (one sigma) before MTTF, we reduce our 50% chance being after a failure by 34.1% points to 15.9%



$$\begin{aligned}1\sigma &= 50 - 34.1 = 15.9 \\2\sigma &= 50 - 34.1 - 13.6 = 2,3 \\3\sigma &= 50 - 34.1 - 13.6 - 2.1 = 0.2 \\4\sigma &= 0.003 \\5\sigma &= 0.00006 \\6\sigma &= 0.0000002\end{aligned}$$



Z-Score

- ◆ How many standard deviations is a point in time away from the MTTF?

$$Z = \frac{t_{replacement} - MTTF}{\sigma}$$

- ◆ So, we could easily take $Z=6$ and solve to

$$t_{replacement} = MTTF - 6 \cdot \sigma$$

- ◆ **But** six sigma is defined as Six Sigma = 3.4 ppm (Parts per Million) and it is

$$P(Z > 4.5) \approx 3.4 \times 10^{-6}$$



Why is 4.5 Sigma called 5 Sigma

- ◆ Motorola developed Six Sigma in the USA 1987
- ◆ Short answer: convention.
- ◆ Longer answer: The mean fluctuates in well designed production processes by 1.5 Sigma over time. So a process once designed at 6 sigma usually produces 4.5 Sigma quality of products. If 4.5 Sigma, which is 3.5ppm defects, is kept over time, it is a six sigma process.
- ◆ To do so, it is required to constantly measure the outcomes of the process and adopt accordingly all quality measures so that the 3.5ppm defects are kept over time.
- ◆ Boom: Six Sigma quality process!



Example with lightbulbs

Bulb	Failure Time (h)
1	950
2	960
3	970
4	980
5	985
6	990
7	995
8	1000
9	1010
10	1020
11	1030
12	1040
13	1050
14	1060
15	1070
16	1080
17	1090
18	1100
19	1110
20	1120

$$MTTF = \frac{950 + 960 + \dots + 1120}{20} = 1030.5$$

$$\sigma = \sqrt{\frac{(950 - 1030.5)^2 + \dots + (1120 - 1030.5)^2}{20}} = 51.06$$

$$t_{replace} = MTTF - 4.5\sigma = 1030.5 - 4.5 \cdot 51.06 = 800.72$$



- ♦ But: Model assumption is standard normal distribution. Weibull is more realistic!



Six Sigma for constant failure rates

- ♦ For constant failure rates with exponential failure distribution six sigma is pointless, since the standard deviation equals the mean, so “sigma-based” preventive replacement has no physical meaning.

$$MTTF = \frac{1}{\lambda} = \sigma$$

$$\begin{aligned} t_{replacement} &= MTTF - 4.5 \cdot \sigma \\ &= MTTF - 4.5 \cdot MTTF \end{aligned}$$

- ♦ Counterintuition: Constant failure rate means very high variance, since in every time interval the same number of components fail.





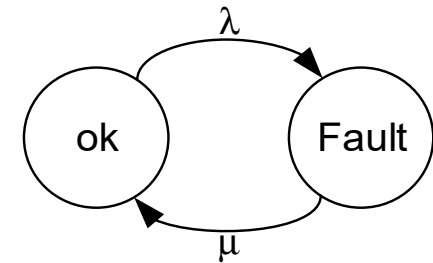
Task: two-way isolation valve

- ◆ We already know, that this is a critical component.
- ◆ MTTF is 120.000h
- ◆ Standard deviation is 10.000h
- ◆ After how many years do we replace to meet Six Sigma?



Repairable Systems

- ◆ Repairable systems have a failure rate λ and a repair rate μ .
- ◆ For the failure rate λ it is $\lambda=1/\text{MTTF}$ and it is $\mu=1/\text{MTTR}$ (Mean Time To Repair)



The *mean time to repair* is the time that a system needs to recover from a fault and its main functions become available again.

- ◆ It is

$$F(t) = \frac{\lambda}{\mu + \lambda} - \frac{\lambda}{\mu + \lambda} e^{-(\mu + \lambda)t}$$

The distribution function

$$\lim_{t \rightarrow \infty} F(t) = \frac{\lambda}{\mu + \lambda}$$

The marginal unavailability
(equals 1 if no repair)

$$\lim_{t \rightarrow \infty} R(t) = \lim_{t \rightarrow \infty} 1 - F(t) = 1 - \frac{\lambda}{\mu + \lambda} = \frac{\mu}{\mu + \lambda}$$

The marginal availability
(equals 0 if no repair)

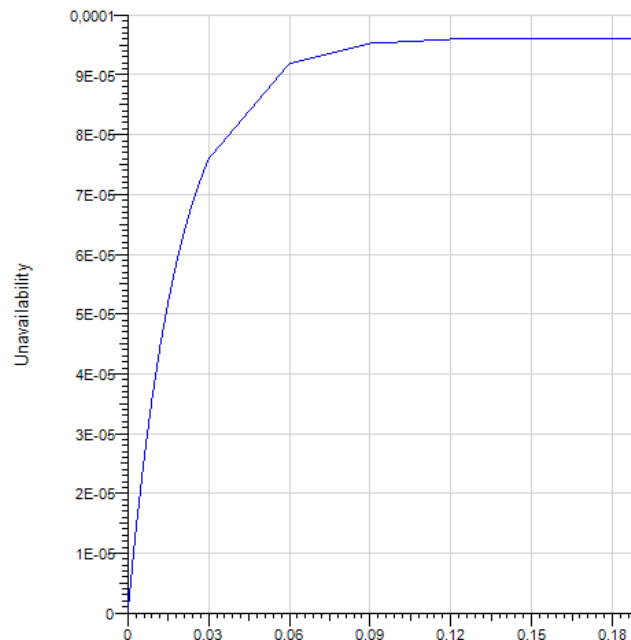
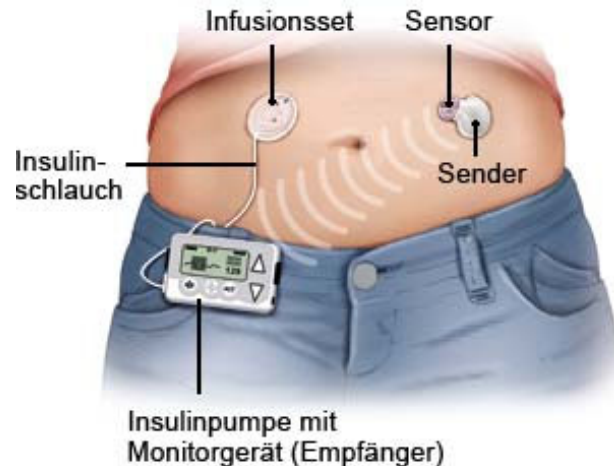


How do we get a repair rate?

- ♦ The repair rate is the inverse of the Mean Time To Repair (MTTR)
- ♦ The MTTR is the mean time between the occurrence of a failure until the function is restored.
- ♦ Usually conservatively estimated



Example: Insulin Pump with Repair Rate



Failure rate $\lambda=0.005$ per year

- ◆ The patient recognizes a failure immediately (by warning signal) and switches to manual injections. He makes an appointment with medical services and receives a new device.

$$\text{MTTR} = 1 \text{ Week} = 1/52 \text{ per year}$$
$$\mu = 1/\text{MTTR} = 52 \text{ per year}$$

- ◆ After approximately 1.5 months, the system has stabilized and the marginal unavailability is reached

$$\frac{\lambda}{\mu + \lambda} = \frac{0.005}{52.005} \approx 1 \cdot 10^{-4}$$

- ◆ As a result, one of 10.000 systems is unavailable.



Six sigma and time to repair

- ◆ Let's realistically assume that our failures are distributed somehow and our time to repair is constant. Then it is

$$t_{replace} = MTTF - MTTR - 4.5\sigma$$



What does that mean for maintenance task planning?

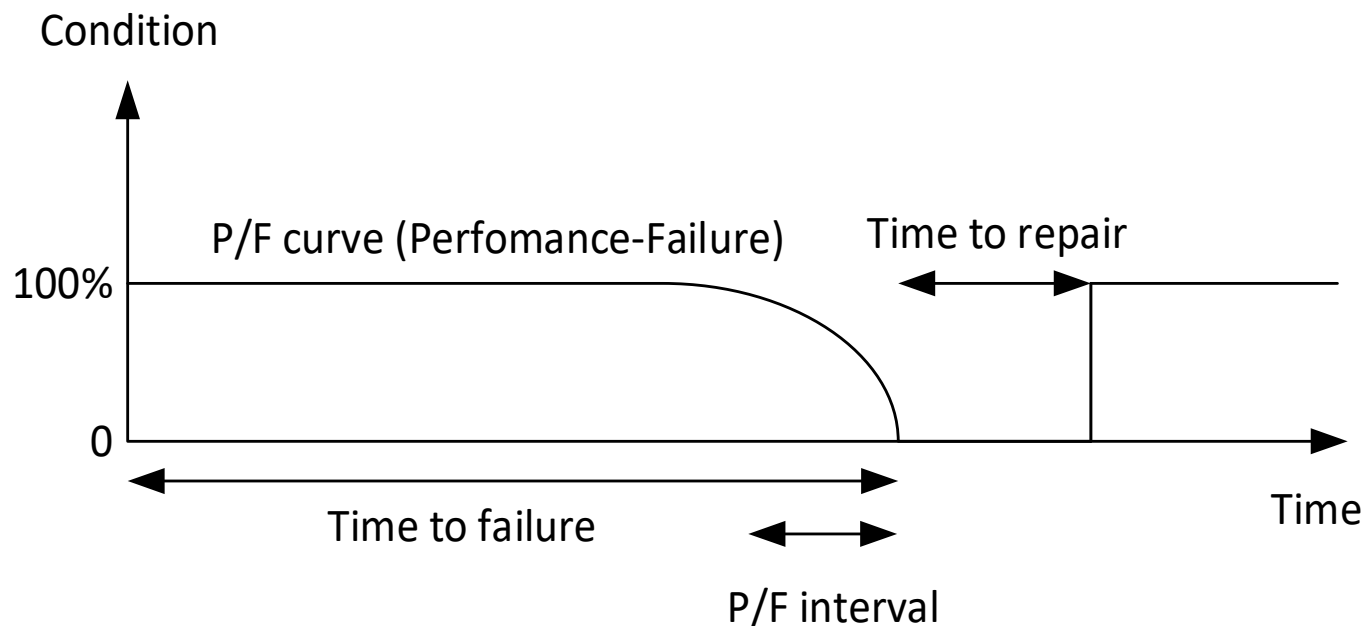
- ◆ If we decide to go for **preventive** maintenance,
 - we need components with a reliable MTTF with low variance and
 - constant failure rates for components are bad for maintenance.

- ◆ If we decide to go for **corrective/predictive** maintenance,
 - we need components that degrade slowly, so that we have time for replacement planning.



Performance-Failure Curve

- ◆ The P/F curve shows the degradation of a system over time.
- ◆ A **potential failure** is a condition that has the potential to lead to a functional failure.
- ◆ A **functional failure** is a condition where the main function of a system is not available in an expected quality.
- ◆ The P/F interval is the time where the main function lost 100% QoS, but is still working somehow.





Examples of components with relevant P/F interval

Component	Potential failure indicators (P)	Functional failure (F)	P/F Interval
Bearing	Vibration Noise	Shaft damage Seizure	Weeks/Months
Gearbox	Vibration Noise Oil debris	Loss of torque Tooth breakage	Weeks/Months
Belts	Cracks Loss of tension Noise	Belt snap Loss of synchronization	Days/Weeks
Drive chains	Increased stretch Lubrication issues Noise	Chain break Jumping teeth	Weeks/Months
Pumps	Leakage Reduced flow Vibration Pressure fluctuations	No flow Seal failure	Days/Months
Valves	Actuation time increases Leaks Torque increases	Stuck at position Leakage	Weeks/Months
Pipes	Wall thickness loss Crack detection Leak	Rupture	Months/years
Electric motors	Current signature anomalies Temperature increases Vibration	Burned windings Bearing seizure Motor stops	Weeks/Months



Components with irrelevant P/F interval

Component	Failure behavior
Electronic ICs	Random failures
Microcontrollers	Sudden death
Fuses	Binary behavior
Light bulbs (filament)	Very short P/F
Relays (sometimes)	Contact welding



Examples of predictive maintenance techniques (1)

- ◆ **Acoustic monitoring**
used to detect internal or external leaks in valves or tubes.
- ◆ **Vibration monitoring and analysis**
in rotating machinery used to detect bearing wear, imbalances, cracks, trend analysis.
- ◆ **Infrared monitoring**
detect loose electronic connections, detect hot spots in motor windings or relay coils.
- ◆ **Oil analysis**
detect gear failures or failures in transformers.
- ◆ **Radiography**
detect subsurface weld flaws or failures in valves.



Examples of predictive maintenance techniques (2)

- ◆ **Magnetic particle inspection or eddy current testing**
detect surface cracks by magnetic fields
- ◆ **Ultrasonic testing**
- ◆ **Liquid penetrant**
use dye to detect surface cracks
- ◆ **Motor current signature analysis**
detect flaws, e.g. by a clamp on testing equipment
- ◆ **Borescope inspections**
internal inspections e.g. for turbines or valves
- ◆ **Diagnostics for motor operated valves**
- ◆ **Diagnostics for air operated valves**



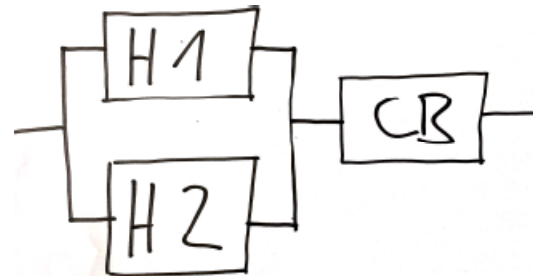
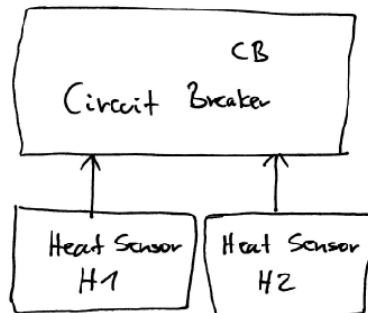
Fine, but..

- ◆ We can decide when a component is run-to-failure and how much cost that involves.
- ◆ We can use 6 Sigma to reduce the failure rate at the cost of planned exchanges of the component.
- ◆ We can use advanced technologies to predict a failure or exchange a component when we measured a bad condition.
- ◆ But if that does not help? → Redundancy



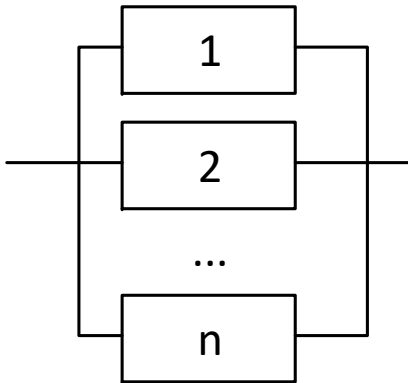
Reliability Block Diagram

- ♦ A *Reliability Block Diagram* (RBDs) is a diagram used to model the reliability when **redundancies** are used.
- ♦ Every RBD can be transformed into a Boolean Fault Tree Model.
- ♦ The RBD is beneficial, since it is focused on the chain of causalities within a system.
- ♦ Modeling paradigm: as long as a path from source to sink is working, the system is available. **Restriction: Component cannot be used multiple times within one diagram!**

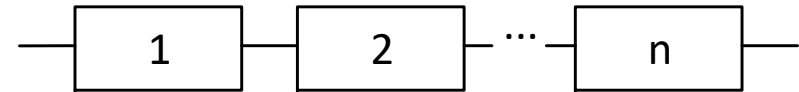




RBD Formula



$$R_{parallel}(t) = 1 - \prod_{i=1}^n (1 - R_i(t))$$

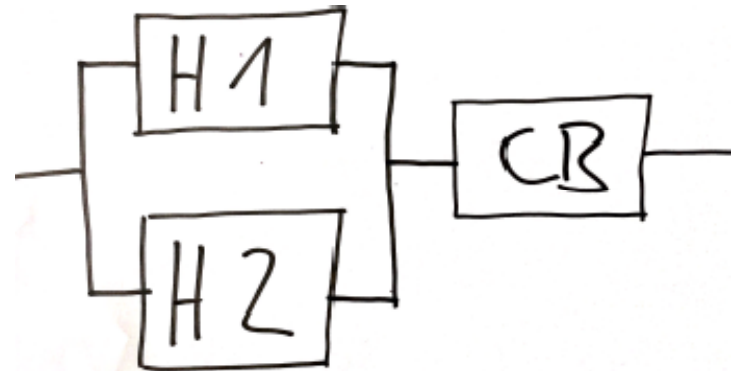


$$R_{serial}(t) = \prod_{i=1}^n R_i(t)$$



Task – How good is redundancy? (20 min)

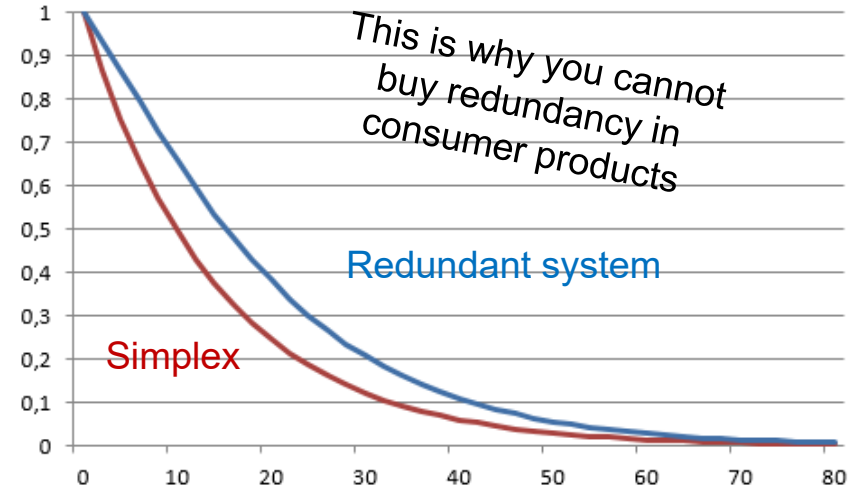
- ♦ Calculate the reliability function $R(t)$ for the constant failure rates of 0.04 for heat sensors H1 and H2 and 0.03 for the circuit breaker CB.
- ♦ How is the probability, that this system is reliable after a time of 20 and 200?
- ♦ How is the probability, that the system without redundant heat sensors is reliable after a time of 20 and 200?
- ♦ How do you judge the result?





Task – How good is redundancy?

- ♦ Calculate the reliability function $R(t)$ for the constant failure rates of 0.04 for heat sensors H1 and H2 and 0.03 for the circuit breaker CB.
- ♦ How is the probability, that this system is reliable after a time of 20 and 200?
- ♦ How is the probability, that the system without redundant heat sensors is reliable after a time of 20 and 200?
- ♦ How do you judge the result?



$$R_{red}(t) = (1 - (1 - R_{H1}(t)) * (1 - R_{H2}(t))) * (R_{CB}(t))$$

$$= (1 - (1 - e^{-0.04t}) * (1 - e^{-0.04t})) * (e^{-0.03t})$$

$$R_{red}(20) = 0.382$$

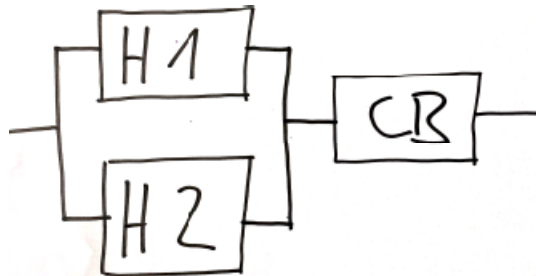
$$R_{red}(200) = 0.0000016$$

$$R_{simpl}(t) = R_{CB}(t) * R_H(t)$$

$$= e^{-0.03t} * e^{-0.04t}$$

$$R_{simpl}(20) = 0.247$$

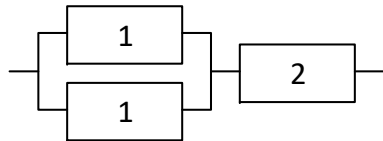
$$R_{simpl}(200) = 0.0000008$$



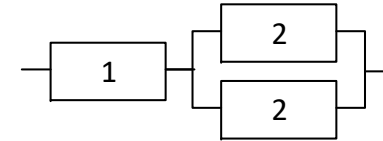


Redundancy Allocation

- ♦ Which components shall be made redundant is an important question in systems engineering
- ♦ In general, it is not possible to make the entire system redundant. For example, you can only fly using one single airplane.

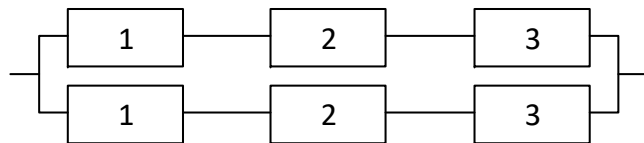


$$R_A = (2R_1 - R_1^2)R_2$$

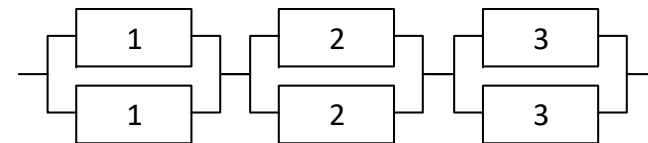


$$R_B = R_1(2R_2 - R_2^2)$$

If it is $R_1 < R_2$, variant A is more reliable. Therefore, it is most beneficial to make the weakest component or the component with the highest failure rate redundant.



$$R_A = 1 - (1 - R_1 R_2 R_3)^2$$



$$R_B = (1 - (1 - R_1)^2) (1 - (1 - R_2)^2) (1 - (1 - R_3)^2)$$

Lo-level redundancy is more reliable than higher level redundancy. System B is more reliable



Types of Redundancies

- ◆ **Passive Redundancy** uses parallel execution and does not detect failures, but masks them and prevents them from propagating through the rest of the system.
- ◆ **Active Redundancy** uses a failure detection mechanism and reacts on a failure e.g. by calculating a result again or doing an action again.
- ◆ It is called **Homogenous Redundancy**, if the redundant modules are identical. Homogeneous redundancy can deal with random hardware faults.
- ◆ It is called **Diverse Redundancy**, if the redundant modules are different but implementing the same specification. Diverse redundancy can cope with systematic faults.
- ◆ A module is called **hot spare**, if the module is active all the time
- ◆ A module is called **cold spare**, if the module is only activated when required

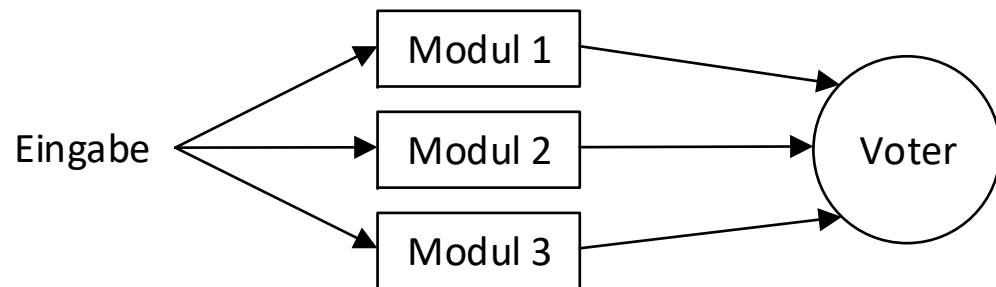


Triple Modular Redundancy

Passive Redundancy

- ♦ *Triple Modular Redundancy* (TMR) uses a voter and selects identical results as the valid results. Redundant modules can be

- Memory
- Processors
- Network connections
- Hard drives
- Bus Systems
- Energy sources

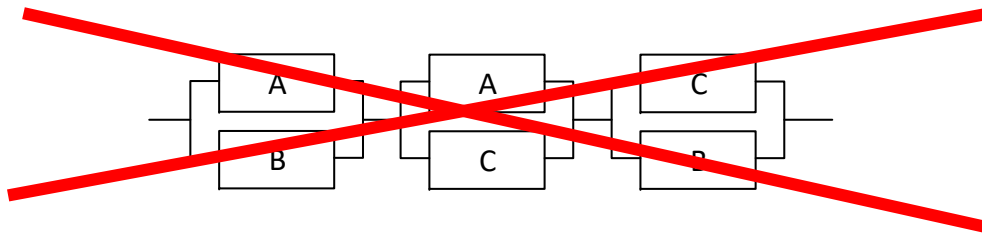


- ♦ Benefit: vast increase of reliability over a short time.
- ♦ Drawback: the voter is a single point of failure and not redundant. Reliability is lower than without TMR if the mission time increases.



TMR Structural Formula

- ♦ TMR cannot be modeled using RBDs, since to do so, modules have to be used twice, which is prohibited in these kind of diagrams.



- ♦ TMR is reliable, if either all three of the modules are working or only one module crashes:

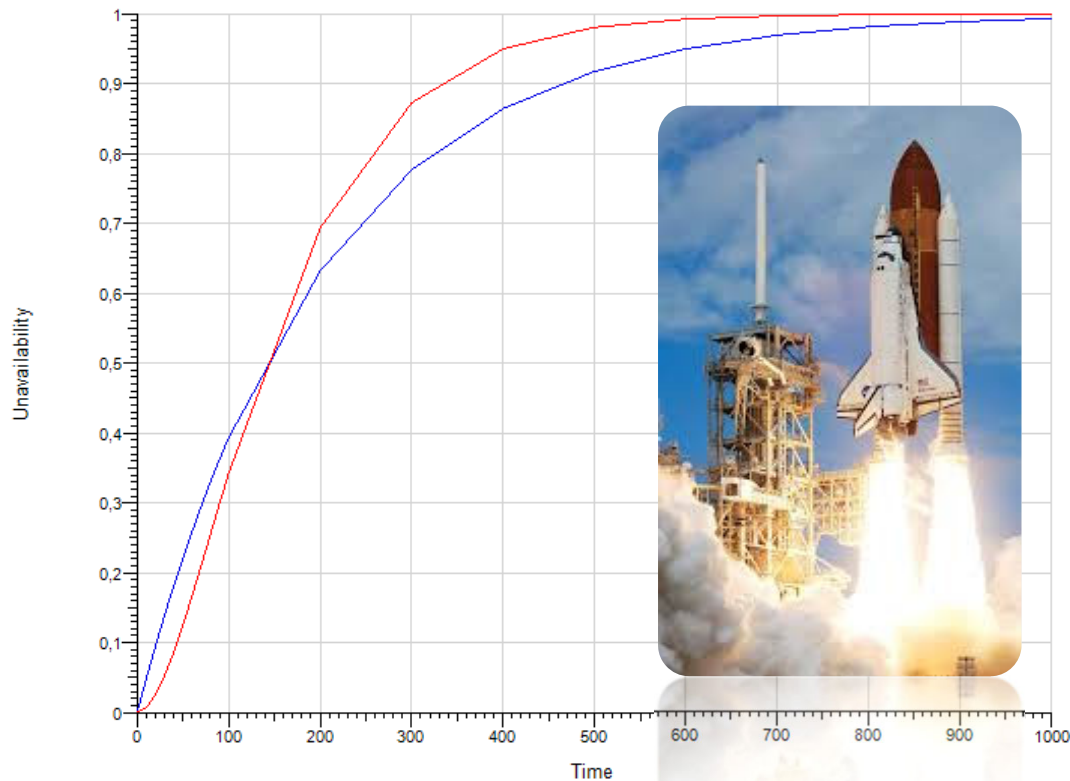
$$R_{TMR} = R_A R_B R_C + (1 - R_A) R_B R_C + R_A (1 - R_B) R_C + R_A R_B (1 - R_C)$$

- ♦ If all modules have the same reliability (e.g. for homogeneous redundancy), it is

$$R_{TMR} = 3R^2 - 2R^3$$



(Un)Reliability of TMR



Blue: without TMR

Rot: with TMR

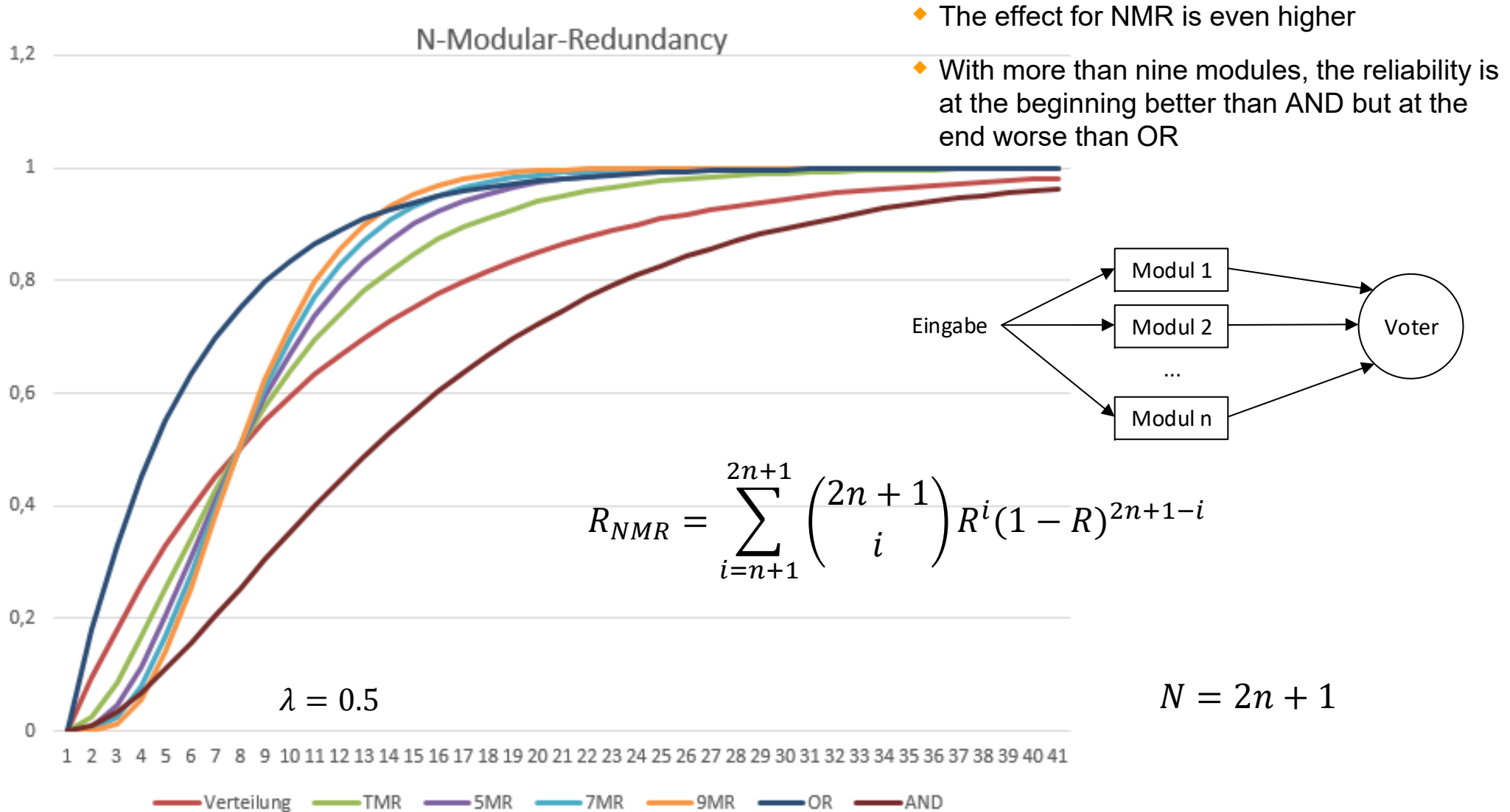
Cuts at 0.5 unreliability. After that, the TMR pattern has a lower reliability.

„After 0.5 is the expectancy value for one module to crash closer to 1. Starting from this point, it is sufficient for the TMR to crash, that one more module crashes. This chance is twice as high, since there are two more modules.“



N-Modular Redundancy

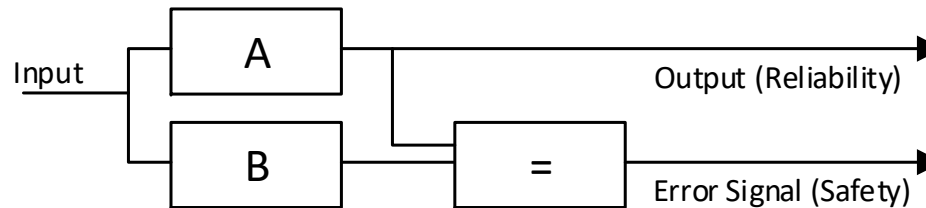
Passive Redundancy





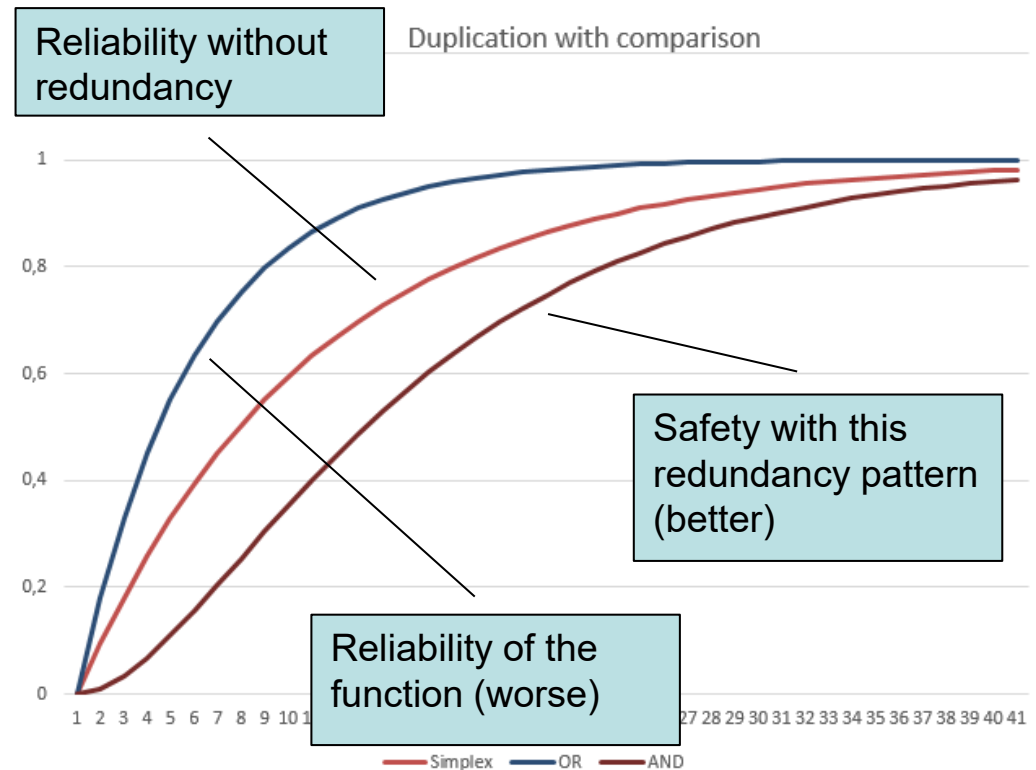
Duplication with comparison

Active Redundancy



- ♦ By duplicating the module, a failure can be detected more reliably. That decreases the reliability of the system, since more failures will be detected than really occur (false negatives)
- ♦ The rate of undetected failures decreases.

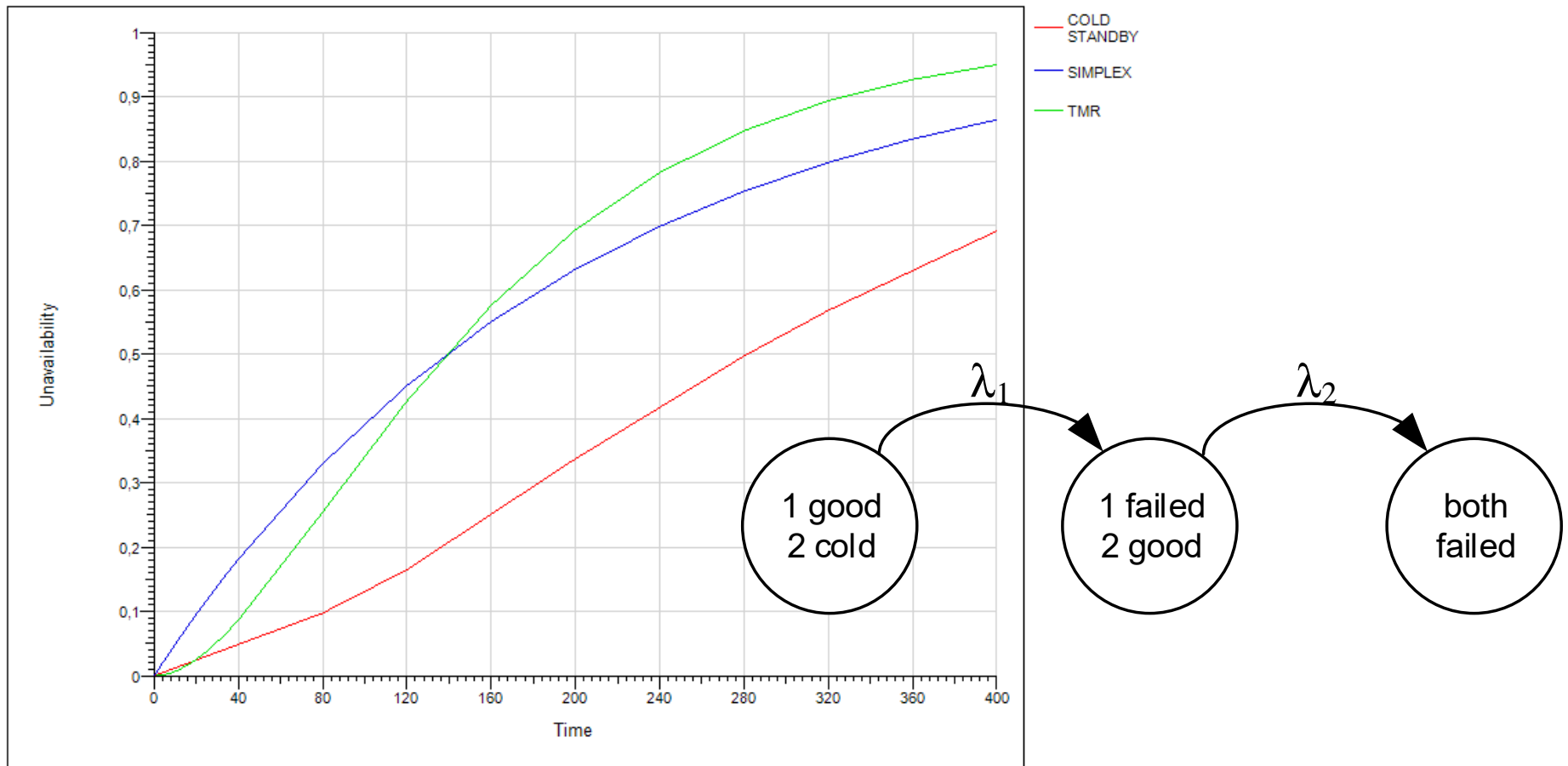
→ Tradeoff between reliability and safety

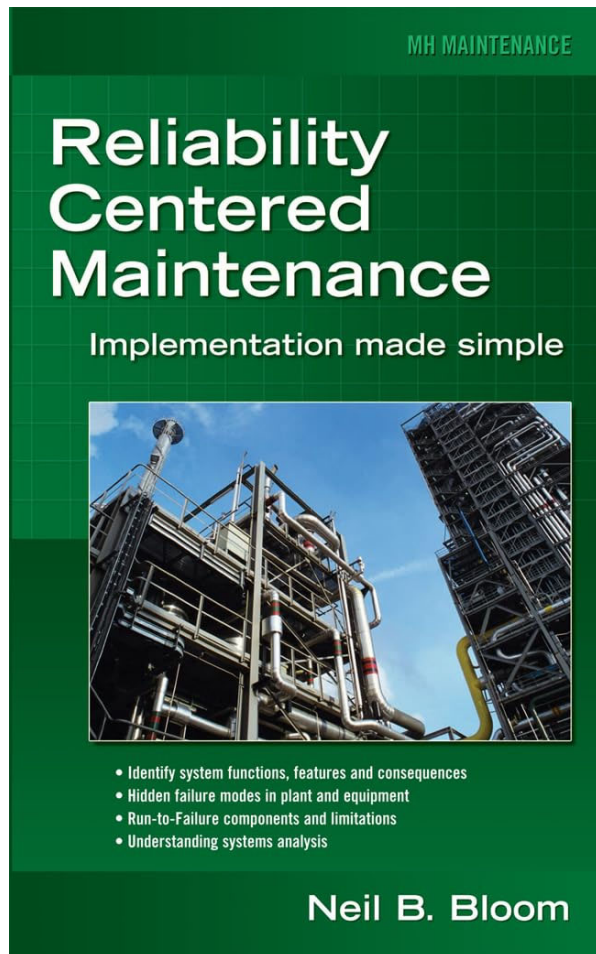




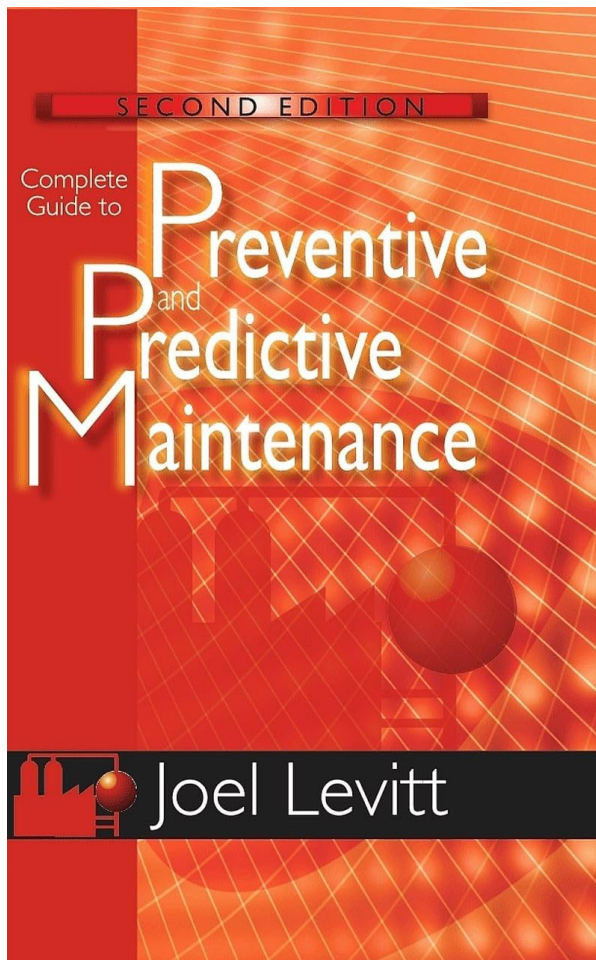
Cold Redundancy

- ♦ Cold spares cannot be modeled using the aforementioned simple modeling techniques, since the lifetime of a cold spare typically starts to decreasing when activated.

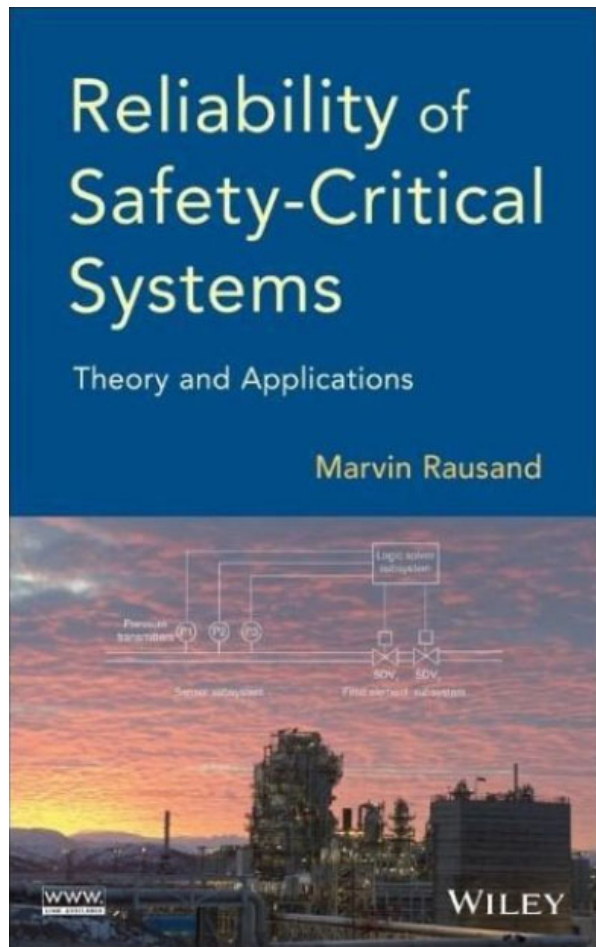




- ◆ Reliability Centered Maintenance (RCM)
- ◆ Neil B. Bloom
- ◆ “A properly implemented and managed RCM program can save millions in unscheduled maintenance and breakdowns. However, many have found the process daunting. Written by an expert with over 30 years of experience, this book introduces innovative approaches to simplify the RCM process such as: single vs. multiple failure analysis, hidden failures analysis, potentially critical components analysis, run-to-failure and the difference between redundant, standby, and backup functions. Included are real life examples of flawed preventive maintenance programs and how they led to disasters that could have easily been avoided. ”



- ◆ Complete Guide to Preventive and Predictive Maintenance
- ◆ Joel Levitt
- ◆ “This book shares the best practices, mistakes, victories, and essential steps for success which the author has gleaned from working with countless organizations. Unlike other books that only focus on the engineering issues (task lists) or management issues (CMMS), this in-depth resource is the first to give true emphasize to the four aspects of success in preventive maintenance systems—engineering, management, economic, and psychological — thereby enabling readers to have a balanced view and understanding of what is happening in their organizations. Additionally, it blends concrete actionable steps and structures with the theory behind the steps. ”



- ◆ Reliability of Safety-Critical Systems
- ◆ The Bible of Reliability Theory
- ◆ Marvin Rausand
- ◆ “Presents the theory and methodology for reliability assessments of safety-critical functions through examples from a wide range of applications
Reliability of Safety-Critical Systems: Theory and Applications provides a comprehensive introduction to reliability assessments of safety-related systems based on electrical, electronic, and programmable electronic (E/E/PE) technology. With a focus on the design and development phases of safety-critical systems, the book presents theory and methods required to document compliance with IEC 61508 and the associated sector-specific standards.”